Package 'mleirt'

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Title IRT MLE with shinyapps

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Description IRT curves with shinyapps.

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`mleirt` *Explore maximum likelihood, EAP and MAP estimation in item response theory*

Description

Launches a Shiny app that shows, with interactive plots, how the estimates of item response theory (IRT) are found: a person's ability by maximum likelihood with the Newton-Raphson method (Rasch model), an item's discrimination and difficulty by maximum likelihood (two- and three-parameter logistic models), and a person's ability by Bayesian expected a posteriori (EAP) and maximum a posteriori (MAP) estimation. The examples follow de Ayala (2009), and the default settings reproduce his published results.

Usage

```
mleirt()
```

Details

All tabs use the logistic item response function

$$P(u = 1 \mid \theta) = c + (1 - c)\frac{\exp(\alpha(\theta - \delta))}{1 + \exp(\alpha(\theta - \delta))}$$

the probability that a person of ability θ answers an item correctly, where δ is the item's difficulty, α its discrimination and c its guessing parameter. The Rasch model fixes $\alpha = 1$ and $c = 0$, the two-parameter logistic model (2PL) fixes $c = 0$, and the three-parameter logistic model (3PL) estimates all three. No scaling constant is used ($D = 1$). The likelihood of a set of responses is the product of the probabilities of the responses given, and the log-likelihood is its logarithm.

The app has four tabs, described in the sections below.

Value

A Shiny app object (`shiny::shinyApp()`). Printing it, which is what happens when you call `mleirt()` at the console, starts the app in the browser or the RStudio viewer; you can also pass it to `shiny::runApp()`.

MLE Rasch

How the Newton-Raphson method finds the maximum likelihood estimate of a person's ability when the item difficulties are known. The test has five items with difficulties -1.90, -0.60, -0.25, 0.30 and 0.45 (de Ayala, 2009, p. 22), and a raw score r is taken to come from answering the first r items correctly: $r = 0$ is the response pattern 00000, $r = 2$ is 11000 and $r = 5$ is 11111.

- **Controls:** the raw score, the starting value of the estimate (-2 to 2) and the iteration to show (1 to 5).
- **Plot:** the log-likelihood of the response pattern for abilities from -6 to 6. For the chosen iteration it shows the estimate the step starts from (red dot), the tangent to the log-likelihood there (blue line) and the estimate the step ends at (green dot), given by $\theta_1 = \theta_0 - LL'(\theta_0)/LL''(\theta_0)$, with the derivatives taken from a smoothing spline of the log-likelihood. The shaded triangle shows the angle between the tangent and the horizontal axis, which is zero at the maximum.
- **Text:** the numbers of the Newton-Raphson step and the angle.

Raising the iteration shows the estimates converging on the maximum of the curve, which reproduces the person locations of de Ayala (2009, Table 2.2, p. 28). For raw scores of 0 and 5 the log-likelihood keeps rising towards minus or plus infinity, so it has no maximum and the steps move away: the maximum likelihood estimate does not exist for zero or perfect scores.

MLE 2PL

The log-likelihood surface of an item's discrimination and difficulty under the 2PL, and where its maximum lies. The app simulates the responses of 50 people, whose abilities are known and spread evenly from -3 to 3, to one item with the true discrimination and difficulty set on the sliders, and draws new responses whenever a true value changes.

- **Controls:** the true discrimination and difficulty (-3 to 3), and two sliders that rotate the plot.
- **Plot:** the log-likelihood of the simulated responses for every combination of discrimination and difficulty from -3 to 3 in steps of 0.1, as a 3-D surface. The red dot marks its highest point: the maximum likelihood estimates of the two parameters. With only 50 responses they differ from the true values by sampling error, and they change with every new draw.

MLE 3PL

The same for the 3PL, to show how the guessing parameter affects the estimation. The app simulates the responses of the same 50 people from the true guessing, discrimination and difficulty set on the sliders, then computes the log-likelihood surface of discrimination and difficulty (-3 to 3 in steps of 0.15) for each of twelve assumed guessing values, 0.05, 0.10, ..., 0.60.

- **Controls:** the true guessing (0.05 to 0.60), discrimination and difficulty, and two sliders that rotate the plot.
- **Plot:** the twelve surfaces drawn over each other in different colours, each with a dot at its highest point; the legend gives the guessing value of each colour. Each surface is drawn on its own vertical scale, so compare where the maxima lie rather than their heights. Computing twelve surfaces takes time, and the plot can take a minute to appear.

EAP-MAP Rasch

Bayesian estimation of a person's ability under the Rasch model, by quadrature. The test has five items with difficulties fixed at -2.155, -0.245, 0.206, 0.984 and 1.211 (de Ayala, 2009, Table 4.4), and raw scores map to response patterns as in the MLE Rasch tab. With prior weights $A(X_r)$ and likelihood $L(X_r)$ at the quadrature points X_r,

$$EAP = \frac{\sum_r X_r L(X_r) A(X_r)}{\sum_r L(X_r) A(X_r)}$$

is the mean of the posterior distribution, and MAP is the quadrature point where the posterior $L(X_r)A(X_r)$ is highest.

- **Controls:** the raw score; the number of quadrature points (10 to 100, spread evenly from -4 to 4); and the mean, standard deviation and skewness of the prior, a skew-normal distribution (`fGarch::dsnorm()`) that is normal when the skewness is 1.
- **First sub-tab, four plots:** the prior weights at the quadrature points; the item trace lines, the probability of each observed response across ability (blue for items answered correctly, red for items answered incorrectly); the likelihood, the product of those probabilities; and the posterior, with lines and values for the EAP (blue) and the MAP (red).
- **Combined sub-tab:** the prior (red), likelihood (blue) and posterior (black, rescaled to be comparable) in one plot, with dashed lines at the prior mean and the posterior mean.

The default settings replicate de Ayala (2009, p. 79). A narrower prior pulls the EAP towards the prior mean, more quadrature points make both estimates more precise, and, unlike the maximum likelihood estimate, EAP and MAP stay finite for raw scores of 0 and 5.

Acknowledgments

The plots in this app were originally developed by Metin Bulus and Wes Bonifay for the irtDemo package (Bulus and Bonifay, 2022), available on CRAN at https://cran.r-project.org/package=irtDemo, and the app keeps their credits on the 3PL tab (Metin Bulus) and the EAP-MAP tab (Metin Bulus and Wes Bonifay). Dimitrios Zacharatos adapted them into this package.

References

Bulus, M., and Bonifay, W. (2022). irtDemo R Package: Pedagogical Interactive Web Applications for Estimation, Scoring, and Multi Dimensionality in Item Response Theory. *Anadolu University Journal of Education Faculty*, 6(1), 92-108. doi:10.34056/aujef.913781

de Ayala, R. J. (2009). *The Theory and Practice of Item Response Theory*. New York: Guilford Press.

Examples

```
if (interactive()) mleirt()
```

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